

Carlos A. Taveras

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Research Interests	Schrödinger Operators, Ergodic Theory, Spectral Theory	
Education	Rice University , <i>Houston, TX</i>	2026–Present
	Ph.D. in Mathematics	
	Advisor: David Damanik	
	Rice University , <i>Houston, TX</i>	2022–2026
	M.S. in Electrical and Computer Engineering	
	Advisors: César A. Uribe and Santiago Segarra	
	Thesis: Efficient Path Planning with Soft Homology Constraints	
	Carnegie Mellon University , <i>Pittsburgh, PA</i>	2017–2022
	B.S. & M.S. in Electrical and Computer Engineering	
Publications	• C.A. Taveras , S. Segarra, C.A. Uribe. “ <i>Gromov-Wasserstein Graph Coarsening</i> ”, IEEE International Conference on Acoustic, Speech, and Signal Processing (ICASSP) 2026, pp. 5961-5965	
	• Y. Xu, C.A. Taveras , L. Beaudrot, C.A. Uribe. “ <i>Identifying common backbones of interactions underlying food webs via non-deterministic alignments</i> ”. IEEE International Conference on Acoustic, Speech, and Signal Processing (ICASSP) 2026, pp. 646-650	
	• C.A. Taveras . “ <i>Efficient Path Planning with Soft Homology Constraints</i> ”. Masters Thesis, 2025	
	• C.A. Taveras , S. Segarra, C.A. Uribe. “ <i>Efficient Path Planning with Soft Homology Constraints</i> ”, IEEE Conference on Control Technology and Applications (CCTA) 2024, pp. 657-662	
	• “ <i>Gromov-Wasserstein Graph Coarsening</i> ” (Poster), IEEE International Conference on Acoustic, Speech and Signal Processing, 2026	
Presentations	• “ <i>Identifying common backbones of interactions underlying food webs via non-deterministic alignments</i> ” (Poster), IEEE International Conference on Acoustic, Speech and Signal Processing, 2026	
	• “ <i>Gromov-Wasserstein Graph Coarsening</i> ” (Talk). NeurIPS 2025 Workshop: New Perspectives in Graph Machine Learning, 2025	
	• “ <i>Efficient Path Planning with Soft Homology Constraints</i> ” (Talk), IEEE Conference on Control Technology and Applications, 2024	
	• Raymond L. Johnson Fellowship	2026
	• Ken Kennedy Institute (KKI) Fellowship	2023
Awards	• Rice Dean’s and ECE STAR Fellowships	2022
	• Graduate Education for Minorities (GEM) Fellowship	2021 & 2022
Conferences and Workshops	• NeurIPS Workshop: New Perspectives in Graph Machine Learning	2025
	• Texas Colloquium on Distributed Learning	2025
	• INFORMS Optimization Society Conference	2024
	• IEEE Conference on Control Technology and Applications	2024
	• ICERM Workshop on Interacting Particle Systems (Link)	2024
	• INFORMS Optimization Society Conference	2023
	• Texas Colloquium on Distributed Learning	2023
	• Konstanz School of Collection Behavior (KSCB)	2023
	• North American School of Information Theory (NASIT)	2023

Teaching	Instructor , <i>Tapia STEM Camps at Rice University</i>	2026
	Led three week-long sessions of <i>Techniques of a Pro Mathematician</i> introducing high school students to undergraduate-level mathematical abstraction and problem solving	
	Tutor , <i>SpiderSmart River Oaks Learning Center</i>	2026
	Tutored several primary and secondary school students in various elementary math subjects	
	Teaching Assistant and Grader , <i>Rice University</i>	2023–2025
	Led recitations, held office hours, and created/graded homework for <i>Random Signals (ELEC 303)</i>, a course on probability and statistics for engineers	
	Teaching Assistant and Grader , <i>Carnegie Mellon University</i>	2021–2022
	Led recitations, held office hours, and graded for <i>Signals and Systems (18-290)</i>, a course on basic properties of discrete/continuous-time linear and time-invariant systems (operators)	
	Held office hours and graded for <i>Fundamentals of Signal Processing (18-491/691)</i>, a course on the basic theory and algorithms of discrete-time signal processing	
Employment	Research Scientist Intern, <i>Adobe Inc.</i>	2022
	Data Science Intern, <i>Lam Research Corp.</i>	2021
	Summer Technology Analyst, <i>Goldman Sachs Inc.</i>	2020
	Undergraduate Validation Intern, <i>Intel Corp.</i>	2019
Service		
	Mentor, <i>Rice GPS Pathways Program</i>	2023–2024